Numbers That Aren't Quantities

AP Statistics · Unit 1 · Topic 1.2 · B: 2026-09-10 · E: 2026-09-11 · TEACHER KEY

CED TETHER

- **SKILL: 2.A** — Describe data presented numerically or graphically.
- ENDURING UNDERSTANDING: VAR-1 Given that variation may be random or not, conclusions are uncertain.
- VAR-1.B Identify variables in a set of data. [Skill 2.A]
- VAR-1.B.1 A variable is a characteristic that changes from one individual to another.
- VAR-1.C Classify types of variables. [Skill 2.A]
- VAR-1.C.1 A categorical variable takes on values that are category names or group labels.
- VAR-1.C.2 A quantitative variable is one that takes on numerical values for a measured or counted quantity.

Essential question: When is a number a quantity, and when is it just a label?

DOK-3 skill: 4.B · **FRQ pattern:** `number-coded-variable-and-meaningless-summary`

DOK rationale: Part (c) asks the student to adjudicate two students' classifications of number-coded variables, justify each with the definition, and predict what specific calculation becomes meaningless under the wrong choice; both classifications of the rating can be defended, so the reasoning, not the label, is what is scored.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Read Maya's and Jordan's lines aloud; do not react.
- Video follow-along from minute 5.
- Board slide back up at minute 33. Circulate; push for the definition, not the label.
- Collect at the door. Score part (c) only, E/P/I. Either position on Rating can earn E.

Students do

- One-sentence first take on the zip-code question before any teaching.
- Video follow-along (individuals, variables, categorical vs quantitative).
- Finish (a)–(c); exit line; turn in.

Questions to ask

- What does a zip code measure? What does it count?
- If I averaged two zip codes, what place would I get?
- Is a 5-star rating twice as good as a 2.5-star rating? Does that matter for your answer?
- Which column would you sort to find the 'biggest'? Which column has no 'biggest'?

Adult role

Solo teacher. Circulate 5 pairs. Never say 'categorical' or 'quantitative' yourself during the finish window; ask what the number measures.

[WARN] WATCH FOR

- 'It's a number so it's quantitative' — the digit-vs-quantity confusion the whole lesson is about.
- Students who get Zip code right by instinct but cannot state the definition (P, not E).
- Listing the newspaper or the school as the individual.

SCORING PART (c) — E / P / I

Expected elements

- **zip-categorical-with-definition** (required) — Says zip code is categorical AND ties it to the definition (measured or counted quantity vs. a label)
- **rating-decided-with-reason** (required) — Takes a position on Rating and defends it with the label-vs-quantity distinction (either position accepted with a correct reason)
- **names-meaningless-calc** (required) — Names a specific calculation (mean, SD, histogram of zip codes) and explains why it has no meaning

E	Zip code categorical with the definition stated, a defended position on Rating, and a specific meaningless calculation explained (e.g. mean zip code is not a place).
P	Two of the three elements, OR all three present but the zip-code reason is only 'it's not really a number' with no definition, OR the meaningless calculation is named but not explained.
I	Agrees with Maya on zip code, or gives no definition-based reasoning, or names no calculation.

Common mistakes

- 'It has digits, so it's quantitative' — confuses how a value is written with what it measures
- Saying zip code is categorical 'because you can't add them' without saying WHY (it measures/counts nothing)
- Treating Rating as obviously quantitative because the quiz app computes an average star rating
- Naming the individuals as 'the newspaper' or 'the school' instead of the six students

Optional #1 — not collected

DOK 2 For each variable, write **C** (categorical) or **Q** (quantitative); for each Q, give the unit: (1) jersey number of a soccer player; (2) number of goals scored this season; (3) shoe size; (4) preferred position (goalkeeper, defender, midfielder, forward); (5) minutes played in the last match.

Key: (1) C — a label, even though numeric. (2) Q, goals. (3) Q, shoe-size units (a measured foot length on a scale); accept C with a reason that sizes are ordered labels. (4) C. (5) Q, minutes.

[STAR] TODAY'S PROBLEM — annotated

A school newspaper surveyed six students for a story about getting to school. Part of the data table is shown.

Two reporters disagree about the numbers in it. **Maya:** "Zip code and Rating are both quantitative — they're numbers, so we can average them." **Jordan:** "They're both categorical. Numbers can be labels."

Getting-to-school survey (minutes; 1–5 stars)

Name	Lunch	ZIP	Min	Stars
Aiden	A	01902	12	4
Bea	B	01905	35	2
Cruz	A	01902	8	5
Dev	C	01904	22	3
Elise	B	01905	41	1
Fiona	A	01901	15	4

FIRST TAKE (5 min)

Is a zip code a quantitative variable? One sentence: yes or no, and the reason you'd give Maya.
Key: Not graded. Most will say 'yes, it's a number'; that is the hook.

Rules callout "VARIABLES (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Who are the individuals in this data set, and what are the four variables?

Key: Individuals: the six students (Aiden, Bea, Cruz, Dev, Elise, Fiona). Variables: lunch period, zip code, commute time, rating of the morning commute.

DOK 2 **(b)** Classify each of the four variables as categorical or quantitative. For each quantitative variable, state its unit.

Key: Lunch period: categorical. Zip code: categorical (numeric labels for places; arithmetic on them is meaningless). Commute time: quantitative, minutes. Rating (1–5 stars): categorical — the numbers are ordered labels for opinions; accept quantitative ONLY if the student states that the numbers are being treated as a measured score and names the unit as 'stars'.

DOK 3 **(c)** Decide who is right about **Zip code** and who is right about **Rating**, using the definition of a quantitative variable in each case. Then name one specific calculation a reader would find meaningless if Maya's classification were used for Zip code, and explain why it is meaningless.

Key: Zip code: Jordan is right. A quantitative variable takes numerical values for a measured or counted quantity (VAR-1.C.2); a zip code measures nothing and counts nothing — it names a place, so it is categorical even though it is written with digits. Rating: Jordan is right under the CED definition (the values 1–5 are category labels for opinions, VAR-1.C.1), though a student who argues that the rating is being used as a measured score and says so explicitly can defend 'quantitative'. Meaningless calculation for zip code under Maya's view: the mean zip code (e.g. averaging 01902 and 01905 to get 01903.5) — it is not a place, and 'larger zip code' does not mean 'more of anything'. Also acceptable: a histogram or standard deviation of zip codes.

EXIT

One thing the video changed about my first take: ____________________